Introduction to Data Science (Fall 2026)

CS 1090a, AC 209a, Stat 109a, or CSCI E-109a

Syllabus; 4/1/2026 SUBJECT TO CHANGE

Instructors

- Pavlos Protopapas (SEAS)
- Kevin Rader (Statistics)

Preceptor

- Chris Gumb (SEAS)

Teaching Fellows

- TF Contacts — the name and Harvard email address of every teaching fellow and course assistant

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Meeting Times

- **Lectures:** Mon & Wed 10:30am-11:45am (Geological Museum 100)
- **Sections:** See registration for times (Rooms TBD)
- **Office Hours:** See the Course Calendar on Canvas.

Course Introduction

Welcome to **CS1090a/AC209a/STAT109a**, also offered by the DCE as **CSCI E-109a**, Introduction to Data Science. The course will focus on the analysis of messy, real-life data to perform predictions and inferences using machine learning and statistical methods.

Material covered will integrate four key facets of an investigation using data:

1. **Data wrangling** - web scraping, data cleaning
2. **Exploratory data analysis** - generating hypotheses and building intuition
3. **Prediction, inference, or statistical learning**
4. **Communication** - justifying decisions and analyzing results

This course is the first in a two-part series. In the spring semester, the curriculum builds upon the content of the fall course, diving deeper into unsupervised learning, deep neural networks for computer vision and language modeling, transformers, and generative models. Students are strongly encouraged to enroll in both the fall and spring courses within the same academic year.

This course fulfills the Quantitative Reasoning with Data (QRD) requirement, with the condition that for the Class of 2029 and later, it must be taken for a letter grade to fulfill this requirement.

Prerequisites

A foundational knowledge of Python programming is required for this course. Specifically, students should be comfortable with the following Python concepts:

- Conditionals, loops, and data structures (e.g., lists, dictionaries, etc.)
- Functions
- File I/O and string parsing

- Classes, methods, attributes, and general OOP principles

Additionally, students should possess an understanding of the following:

- Random variables and common probability distributions (e.g., normal and binomial)
- Basic concepts in probability (e.g., independence, joint and conditional probabilities)
- Calculus at an introductory level (*Note: Multivariable calculus is not required*)

If you are unsure about your preparation, HW0 will help you assess readiness for the course.

Topics

- Web scraping
- Manipulating tabular data: Pandas
- Exploratory data analysis (EDA)
- kNN & linear regression
- Multiple & polynomial regression
- Model selection & cross-validation
- Regularization (LASSO and Ridge)
- Maximum likelihood estimation (MLE)
- Bootstrap, confidence intervals, & hypothesis testing
- Bayesian inference, sampling, and hierarchical models
- High dimensionality & principal component analysis (PCA)
- Missing data & imputation
- Classification & logistic regression
- Decision trees
- Ensemble methods:
  - Bagging
  - Random forests
  - Boosting
  - Mixture of experts
  - Meta-models and model stacking

Differences Between CS1090A & AC209A

- **Assignments:** Homework assignments for 209A students may have additional components.
- **Midterms:** The two midterms will differ slightly between 1090A and 209A students.
- **Projects:** Project groups with one or more 209A students will perform some self-directed outside reading to inform their projects. The goal is to make use of some approach or method not explicitly covered in class and communicate an understanding of the method(s) and its applicability to the problem of focus through the final report and presentation.

Course Components

Lectures and sections will be live-streamed for Extension School students through the 'Zoom' section on Canvas.

Lecture recordings will be made available to all registered students within 24 hours and can be accessed through the 'Course Videos' section on Canvas. Extension students will also have access to section recordings.

Lecture

The class meets for lectures twice a week (Mon & Wed). Attendance and participation are an essential part of learning the material in this course.

On-campus students are expected to attend all lectures and sections in person. Extension students are not required to attend live lectures but are responsible for all course content and deadlines.

Readings

Short readings may be assigned before or after some lectures to provide context, background, or deeper discussion of selected topics. These may include brief reviews, articles, or technical notes related to the course material.

Readings are **not directly graded**, but students are expected to use them to support their understanding of the lectures.

Quizzes

There will be **in-class quizzes**, scheduled as shown on the course schedule once finalized. Each quiz will take approximately **15 minutes** and will be administered during class time.

The **lowest two quiz scores will be dropped**. This drop policy is intended to provide maximum flexibility for illnesses, travel, or other unforeseen circumstances. **No additional quiz drops, make-ups, or extensions will be granted beyond this policy.**

Resources you may use during a quiz. You may consult the lecture PDFs and the section materials, opened through Ed. These are the only permitted resources. You may not use:

- your own notes, whether handwritten or printed;
- notebooks opened in VS Code, Cursor, Jupyter, or any other code editor;
- AI tools, search engines, or any other website, application, or file.

For **"select all that apply" questions**, **partial credit will not be awarded**. Only fully correct selections will receive credit.

Extension students will take quizzes online outside of class time or via an alternative mechanism to be communicated by the instructional staff.

Homework

- **HW0** will be released before the start of the semester. It serves to evaluate students' preparedness for the course, covering the prerequisites listed above. Students finding this material too challenging should consider registering for the course in a future semester, after having spent more time with the prerequisites.
- **HW0 must be completed individually.** It is graded on **completion** rather than correctness: a genuine attempt at every part earns full credit, and a wrong answer costs nothing. What students receive back is a per-question diagnostic of their own preparation, which is the point of the assignment.
- Because HW0 is released before the semester begins, the course computing environment may not yet be available. HW0 includes instructions for setting up Python locally, and is distributed as a PDF as well as a notebook so students can begin reading before they have installed anything.
- After HW0, there will be **5 homework assignments**, each due roughly **two weeks** after being assigned.
- For **these** assignments (HW1–HW5, not HW0), students have the option to collaborate and submit their work in pairs. Please consult the collaboration policy below.

Collaboration rules apply to homework assignments only; quizzes, midterms, and projects follow separate policies.

Midterms

The course will have **two midterm exams** taking place on the evenings of 10/21 and 12/2.

On-campus students are required to take midterms in person. On-campus students may not take the exam online.

Extension School students will take the midterms online. Multiple time slots will be provided to accommodate different time zones. The online exam for Extension students may differ from the in-person exam.

Midterms are **closed-book**. Students may bring **one two-page cheatsheet**. The cheatsheet may be either **printed or handwritten**. Two pages may be used as **one double-sided page**.

Project

Students may propose their own project topic or adapt an example provided by the teaching staff. Students self-assemble into groups and submit their preferences from an approved set of project options.

Each group will then be assigned a project and a teaching staff mentor, who will provide guidance and feedback throughout the project.

Milestones

0. Project proposal
1. Preferences & group formation
2. Data check, refined problem statement, and course-of-action
3. Data processing, EDA, and baseline analysis
4. Final Presentations & Report

Milestones **2 and 3** will be evaluated through a **live presentation to the group's assigned Teaching Fellow**, in addition to submitted materials.

Detailed expectations, guidelines, and rubrics for each milestone will be provided during the first few weeks of the course.

Projects are group-based and must be completed by teams of **3-5** students. Pair and solo projects are not permitted.

Grading

Final grades for the course will be computed using the following weights:

Assignment	Final Grade Weight
Homework 0	1%
Homework: 1-5	29%
Lecture Quizzes	8%
Midterms	37%
Final Project	25%
Total	**100%**

Grading weights apply to both 1090A and 209A unless otherwise stated.

This course uses the grading system outlined in the FAS Student Handbook:

A, A- Earned by work whose excellent quality indicates a full mastery of the subject and, in the case of the grade of A, is of extraordinary distinction.

B+, B, B- Earned by work that indicates a good comprehension of the course material, a good command of the skills needed to work with the course material, and the student's full engagement with the course requirements and activities.

C+, C, C- Earned by work that indicates an adequate and satisfactory comprehension of the course material and the skills needed to work with the course material and that indicates the student has met the basic requirements for completing assigned work and participating in class activities.

D+, D, D- Earned by work that is unsatisfactory but that indicates some minimal command of the course materials and some minimal participation in class activities that is worthy of course credit toward the degree.

E Earned by work that is unsatisfactory and unworthy of course credit toward the degree.

Notes: - Numerical scores in the class will be converted to letter grades at the end of the course by the instructors. - Project grades are assigned at the group level, except in cases of documented and significant imbalance in contribution. - Regrade requests must be submitted via Gradescope within one week of grade release and must clearly identify a grading error. Requests for re-evaluation of subjective judgment will not be considered.

Attendance

Attendance is expected in this course. We will not formally track attendance in this course. This reflects our belief that attendance should be driven by its value to your learning.

Quizzes will be administered **during class time** and cannot be taken outside the scheduled class time.

Based on past offerings of the course, there is a strong correlation between performance and attendance. Students are therefore strongly encouraged to attend regularly.

The instructors strongly believe that regular attendance is essential to the success of the course and significantly improves the experience for all students and the instructional staff through stronger discussion, engagement, and feedback.

Serious Illnesses and Exceptional Circumstances

The course policies on dropped quizzes, late days, and deadlines are intended to fully cover routine and foreseeable situations, including short-term illnesses that do not involve hospitalization or prolonged incapacity, as well as interviews, travel, family events, weddings, and similar scheduling conflicts.

For **serious illnesses or exceptional circumstances** that go beyond these cases, students must **communicate with the instructional staff as early as possible**. Waiting until after a deadline or assessment has passed may limit our ability to provide accommodations.

Requests for accommodations due to serious illness or exceptional circumstances will be **evaluated on a case-by-case basis** and may require appropriate documentation. There is **no automatic extension or exemption**, and accommodations are not guaranteed.

The instructional staff reserves the right to determine what constitutes a serious or exceptional circumstance and what, if any, accommodation is appropriate.

Resources

Recommended Textbook

The book for the course is *An Introduction to Statistical Learning* and is available at statlearning.com.

Software

Our primary software tools will be **Python 3**, **Jupyter notebooks**, and standard third-party Python libraries.

SEAS provides the **FASOnDemand** service, accessible through Canvas. This is a remote Python environment where students can run Jupyter notebooks. The environment includes all packages used in the course and requires no local installation.

Late Work Policy

All students are allocated a maximum of **4 late days** to use throughout the semester. **At most 1 late day can be applied to any single homework assignment.**

If a student has exhausted all their late days, late homework will not be accepted unless there is a medical (if accompanied by a doctor's note) or other official, University-excused reasons. There is no need to ask before using one of your late days. This late day policy is designed to provide maximum flexibility for unforeseen circumstances, including illness. Therefore, no additional late days will be granted beyond this allocation.

Late days cannot be applied to **HW0**, quizzes, midterms, or project milestones.

Academic Integrity

We expect you to adhere to the Harvard Honor Code at all times. Failure to adhere to the honor code and our policies may result in serious penalties, up to and including automatic failure in the course and reference to the ad board.

DCE Academic Integrity Policy

If you are an Extension School student, you are responsible for understanding Harvard Extension School policies on academic integrity (https://extension.harvard.edu/for-students/student-policies-conduct/academic-integrity/) and how to use sources responsibly. Stated most broadly, academic integrity means that all course work submitted, whether a draft or a final version of a paper, project, take-home exam, online exam, computer program, oral presentation, or lab report, must be your own words and ideas, or the sources must be clearly acknowledged. The potential outcomes for violations of academic integrity are serious and ordinarily include all of the following: required withdrawal (RQ), which means a failing grade in the course (with no refund), the suspension of registration privileges, and a notation on your transcript.

Using sources responsibly (https://extension.harvard.edu/for-students/support-and-services/using-sources-effectively-and-responsibly/) is an essential part of your Harvard education. We provide additional information about our expectations regarding academic integrity on our website. We invite you to review that information and to check your understanding of academic citation rules by completing two free online 15-minute tutorials that are also available on our site. (The tutorials are anonymous open-learning tools.)

Student Collaboration

If you work with a partner on an assignment make sure both parties solve all the problems. Do *not* divide and conquer. You are expected to be intellectually honest and give credit where credit is due. In particular:

- if you work with a fellow student and want to submit the same notebook you need to form a group prior to the submission. Details in the assignment. Not all assignments will permit group submissions.
- you need to write your solutions entirely on your own or with your collaborator.
- if you worked with a fellow student on a paired assignment but decide in the end to submit different notebooks individually, include the name of the other student as a comment at the top of your notebook.
- you are welcome to take ideas from code presented in sections, lecture, or sections but you will need to change it, adapt it to your style, and ultimately write your own. **Simply copying verbatim will rarely be successful**.
- if you use code found on the internet, books, or other sources you need to cite those sources.
- you should not view any written materials or code created by other students for the same assignment.
- you may not provide or make available solutions to individuals who take or may take this course in the future. If you are using a remote git repository such as GitHub to work on your assignments **you must make it private.**

Use of AI Models

Purpose of Policy: This policy outlines the acceptable use of AI models, including but not limited to ChatGPT, in completing assignments for this course.

Policy Guidelines:

1. **Original Work:** Students are expected to complete assignments using their original thoughts and interpretations. AI models can be used to help understand concepts, or learn about different perspectives, but they should not write or complete assignments for students.
2. **Collaboration with AI:** Students may use AI models for brainstorming or generating preliminary ideas, but the final work submitted must be substantially their own. Students should be able to explain their reasoning, logic, and conclusions without relying on the model's output.
3. **Contextual Restrictions:** There may be specific course components (e.g. quizzes, midterms, etc.) where the use of AI models is entirely prohibited. These restrictions will be clearly stated in those contexts.
4. **Ethical Considerations:** Students are encouraged to approach the use of AI with ethical considerations in mind, including issues related to privacy, bias, and authenticity.

Consequences for Non-Compliance: Failure to adhere to this policy may result in academic penalties as outlined in the course's academic integrity policy.

Questions and Clarifications: If students have questions about the appropriate use of AI models in an assignment, they should consult the course instructors or teaching assistants before proceeding.

Disclosure of AI use is not required.

Please refer to the University's policy for further information.

AI Policy: TLDR

- **Correct Use**: Using AI to support deeper engagement with course content and the development of one's own skills and understanding.
- **Incorrect Use:** Using AI to *replace* that engagement and development (e.g., to generate work or reasoning that the student cannot independently explain and reproduce)

Accommodations for Students with Disabilities

Harvard students needing academic adjustments or accommodations because of a documented disability must present their Accommodation Action Plan from the Disability Access Office (DAO) and speak with the Course Heads as early as possible in the semester. Failure to do so may result in the Course Heads' inability to respond in a timely manner. All discussions will remain confidential, although Faculty are invited to contact DAO to discuss appropriate implementation.

Harvard Extension School is committed to providing an inclusive, accessible academic community for students with disabilities and chronic health conditions. The Accessibility Services Office (ASO) https://extension.harvard.edu/student-experience/accessibility-services/ offers accommodations and supports to students with documented disabilities. If you have a need for accommodations or adjustments in your course, please contact the Accessibility Services Office by email or by phone.

Diversity and Inclusion Statement

As educators, we aim to build a diverse, inclusive, and representative community offering opportunities in data science to everyone. We will encourage learning that advances ethical data science, exposes bias in the ways data & data science can be (and all too frequently is) used, and advances research into fair and responsible data science.

We need your help to create a learning environment that supports a diversity of thoughts, perspectives, and experiences, and honors your identities (including but not limited to race, gender, class, sexuality, religion, ability, etc.) To help accomplish this:

- If you have a name and/or set of pronouns that differ from those in your official Harvard records, please let us know!

- If you feel like your performance in the class is being impacted by your experiences outside of class, please do not hesitate to come and talk with us. We want to be a resource for you. Remember that you can also submit anonymous feedback (which will lead to us making a general announcement to the class, if necessary, to address your concerns). If you prefer to speak with someone outside of the course, you may find helpful resources at the Harvard Office of Community and Campus Life.
- We (like many people) are still learning about diverse perspectives and identities. If something was said in class (by anyone) that made you feel uncomfortable, please talk to us about it.
- As a participant in course discussions, you are expected to respect your classmates' diverse backgrounds and perspectives.

Our course will discuss diversity, inclusion, and ethics in data science. Please contact us (in person or electronically) or submit anonymous feedback if you have any suggestions for how we can improve.

For additional resources, guidance, and support related to diversity and inclusion, please refer to the Harvard Office of Community and Campus Life.

Auditing

To request to audit the course, send an email to chrisgumb@seas.harvard.edu with your HUID (required) and a statement of agreement to the terms below. **Note:** Please make sure you are not currently enrolled in the course when you send your request. You can't be added as an auditor in Canvas if you are currently listed there as an enrollee.

All auditors must agree to abide by the following rules:

- Auditors must attend class in person. This is a Harvard policy. Auditors who do not confirm their presence during the first week of in-class instruction will lose course access.
- Auditors are held to the same standard of academic honesty as our registered students. Please do not share homeworks or solutions with anyone. Violations will be reported to the Harvard Administrative Board.
- Auditors are not permitted to take the course for credit in the future.
- Auditors should **not** submit HWs or midterms, or participate in projects.
- Auditors should refrain from using any course and TF resources that are designed for our registered students like Ed, FASOnDemand, and office hours.

Frequently Asked Questions (FAQ)

Q1. Is attendance mandatory?

For **on-campus students**, attendance is expected and strongly encouraged, but it is not formally tracked. Quizzes are administered during class time, and regular attendance is strongly associated with better performance.

Extension School students are not required to attend live lectures but remain responsible for all course content, quizzes, and deadlines.

Q2. Do Extension School students need to attend live lectures?

No. Extension students are not required to attend live lectures but are responsible for all course content, deadlines, quizzes, and exams.

Q3. How do Extension School students take quizzes?

Extension students will take quizzes online or through an alternative mechanism communicated by the instructional staff. Quizzes must be completed during these scheduled quiz windows.

Q4. Can I use my own notes or AI tools during a quiz?

No. During a quiz you may consult only the lecture PDFs and the section materials, opened through Ed. Your own notes are not allowed, whether handwritten or printed. AI tools, search engines, code editors such as VS Code or Cursor, and any other website, application, or file are not allowed either.

Q5. Can I miss a quiz due to illness, travel, or an interview?

The lowest two quiz scores are dropped to cover routine and foreseeable situations. No additional quiz make-ups or extensions will be granted beyond this policy.

Q6. What counts as a serious illness or exceptional circumstance?

Serious illnesses or circumstances are those involving hospitalization, prolonged incapacity, or similarly severe situations. These must be communicated to the instructional staff as early as possible and may require documentation.

Q7. Are midterms open-book?

No. Midterms are closed-book. Students may bring one two-page cheatsheet (either two single-sided pages or one double-sided page), printed or handwritten.

Q8. Can on-campus students take midterms online?

No. On-campus students are required to take midterms in person.

Q9. Can I work alone on the course project?

No. Projects must be completed in groups of 3-5 students. Solo projects are not permitted.

Q10. Are project grades individual or group-based?

Project grades are assigned at the group level, except in cases of documented and significant imbalance in contribution.

Q11. Can I use AI tools for homework or projects?

AI tools may be used for brainstorming, clarification, or debugging, but they may not replace your own reasoning, implementation, or writing. Students must be able to explain their work independently.

Q12. Do I need to disclose AI use?

Disclosure is not required.

Q13. Can I use late days on HW0, quizzes, midterms, or project milestones?

No. Late days apply only to regular homework assignments.

Q13a. Can I work with a partner on HW0?

No. HW0 must be completed individually. It is a check on your own preparation, and it cannot tell you anything useful about that if the work is not yours. Pair submissions are permitted on HW1–HW5 only.

Q13b. How is HW0 graded?

On completion. A genuine attempt at every part earns full credit; a wrong answer costs nothing. You also receive a per-question diagnostic — solid, shaky, or not attempted — which does not affect your score and is the actual point of the assignment.

Q14. Are there fixed numerical cutoffs for letter grades?

No. Numerical scores are converted to letter grades at the end of the course by the instructors, based on overall performance and consistent application of course standards. Individual grade negotiations will not be considered.

Q15. What if something is unclear or not covered here?

When in doubt, ask the instructional staff **before** submitting work or missing an assessment. Clarifications given in writing by the staff take precedence.